Critical rainfall intensity of unsaturated slopes: how much do uncoupled, passive-air analyses overestimate it?

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Abstract

Physically based rainfall thresholds for unsaturated slopes usually come from uncoupled seepage analyses with atmospheric pore air followed by stability analyses at successive instants. We quantify how these hypotheses bias the rainfall that triggers failure. A coupled formulation with displacement, pore-water and pore-air pressures (Bishop's effective stress, Mohr–Coulomb plasticity) is implemented in the open-source library NeoPZ and validated against the Liakopoulos drainage test; the factor of safety (FS) is obtained by strength reduction with frozen pressures. Since values below one are not physical states, models are compared by the critical intensity I_c of a 24-h rainfall, the smallest that brings the FS to one, and by the time to failure. For the slope of Rahardjo et al. (2007) with $k_s = 36$ mm/h, the uncoupled, passive-air model gives $I_c = 48$ mm/h ($1.3k_s$); the coupled model with active air gives 31 mm/h ($0.9k_s$) with an associated flow rule and 26 mm/h ($0.7k_s$) without dilatancy. The usual hypotheses thus overestimate the failure-triggering rainfall by 54–86% and delay failure by up to 10 h. Their effects are opposite: trapped air halves I_c, coupling raises it by 4–63%, and the air dominates. Once the surface ponds, the air below the wetting front is compressed to 36–38 kPa; less water infiltrates and the matric suction increases, but the pore-water pressure follows the air pressure and the effective stress drops, so that a low-permeability slope deemed safe under short rainfalls reaches limit equilibrium. Dilatancy shifts I_c by about 20% and decides whether the transient collapses. The bias persists with $\chi = S_e$ and mesh refinement and vanishes when I_c is well below k_s.

Keywords: unsaturated slope, rainfall infiltration, pore-air pressure, hydro-mechanical coupling, dilatancy, critical rainfall intensity, strength reduction method

1. Introduction

Rainfall-induced landslides in residual and colluvial soils are usually triggered by the loss of matric suction and the rise of pore-water pressure in a shallow unsaturated zone. Warning systems rely on rainfall thresholds, empirical (Caine, 1980; Guzzetti et al., 2008) or physically based, the latter obtained by transient seepage analyses followed by a stability analysis at successive instants. In engineering practice this chain is a seepage code solving Richards' equation on a rigid skeleton with the pore air at atmospheric pressure (SEEP/W, HYDRUS, the hydraulic module of PLAXIS), followed by limit equilibrium (SLOPE/W) or by strength reduction with the pore pressures of each instant (Ng and Shi, 1998; Rahardjo et al., 2007; Tang et al., 2018; Pradhan et al., 2022).

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The parametric study of Rahardjo et al. (2007) is a reference of this approach. For a homogeneous slope they concluded that the saturated permeability k_s controls the response: there is a limiting intensity above which the minimum factor of safety (FS) no longer decreases, a lower FS plateau of about 0.81–0.82, slopes with $k_s \leq 10^{-6}$ m/s do not fail in rainfalls of up to 24 h, and the slope angle, height and initial water table are secondary. Three hypotheses of that chain are rarely questioned together.

1. *Atmospheric pore air.* When the ground surface ponds, the air between the wetting front and the water table can only leave through the surface and is compressed by the infiltrating water. Laboratory columns (Touma and Vauclin, 1986) and two-phase flow analyses of slopes (Cho, 2016) show that the air delays the wetting front and that its pressure lowers the FS; these slope analyses, however, typically solve the flow on a rigid skeleton and evaluate the stability by limit equilibrium.
2. *Uncoupled seepage.* The volumetric strain of the skeleton changes the pore pressures. Coupled analyses can give smaller FS than uncoupled ones (Qi and Vanapalli, 2015) and have been used to back-analyse failures (Yang et al., 2017), but the comparisons are made with the minimum FS, not with the rainfall that triggers the failure.
3. *Flow rule.* The coupled models that predict failure use non-dilatant soils (Wang et al., 2018; Liu and Wang, 2021), while the strength reduction method is usually applied with an associated flow rule. Dilatancy generates suction during plastic shearing and can arrest a coupled failure; none of the studies we reviewed varies it.

This paper quantifies how these three hypotheses bias the critical rainfall intensity of an unsaturated slope, that is, the smallest intensity of a rainfall of given duration that brings the slope to failure. A three-field formulation (displacement, pore-water and pore-air pressures) is implemented in the open-source finite element library NeoPZ (Devloo, 1997) and validated against the drainage test of Liakopoulos (1965). Since a factor of safety below one is not a physical state, the models are compared by the critical intensity I_c and the time to failure, not by the minimum FS. The slope of Rahardjo et al. (2007) is the reference case: the four combinations of coupling and air model isolate the contribution of each hypothesis, and the effect of dilatancy, of the soil permeability, of the Bishop parameter, of the confinement of the air and of the mesh is then assessed.

2. Formulation and implementation

2.1. Governing equations

The soil is a deformable porous medium with two immiscible fluids, water and air, under isothermal conditions; grains and water are nearly incompressible and the air is an ideal gas; dissolved air and vapour are neglected. Small strains and quasi-static equilibrium are assumed. With tension positive and pore pressures positive in compression, the total stress is

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' - p_B\,\mathbf{I}, \qquad p_B = \chi\,p_w + (1-\chi)\,p_a, \tag{1}$$

where $\boldsymbol{\sigma}'$ is Bishop's effective stress (Bishop, 1959) and χ the Bishop parameter. The unknowns are the displacement $\mathbf{u}$ and the gauge pressures p_w and p_a; the capillary pressure is

$p_c = p_a - p_w$. The balance equations of momentum, water mass and air mass are

$$\nabla \cdot \boldsymbol{\sigma} - \gamma\, \mathbf{e}_y = \mathbf{0}, \tag{2}$$

$$\frac{nS_w}{K_w}\,\dot{p}_w + n\,\dot{S}_w + S_w\,\dot{\varepsilon}_v + \nabla \cdot \mathbf{q}_w = 0, \qquad \frac{nS_a}{p_a + p_{atm}}\,\dot{p}_a - n\,\dot{S}_w + S_a\,\dot{\varepsilon}_v + \nabla \cdot \mathbf{q}_a = 0, \tag{3}$$

with $S_a = 1 - S_w$, $S_w = S_w(p_c)$, ε_v the volumetric strain (positive in expansion), γ the total unit weight, $\mathbf{e}_y$ the upward unit vector, a unit Biot coefficient and Darcy's law

$$\mathbf{q}_w = -\frac{k_{rw}}{\gamma_w}\,\mathbf{K}\,(\nabla p_w + \gamma_w \mathbf{e}_y)\,, \qquad \mathbf{q}_a = -\frac{k_{ra}}{\gamma_w}\frac{\mu_w}{\mu_a}\,\mathbf{K}\,(\nabla p_a + \gamma_a \mathbf{e}_y)\,, \tag{4}$$

where $\mathbf{K}$ is the saturated conductivity tensor, k_{rw} and k_{ra} the relative permeabilities and $\mu_w/\mu_a = 55.6$. The retention curve is that of Fredlund and Xing (1994) with the correction factor $C(\psi) = 1$, $S_w = [\ln(e + (p_c/a)^n)]^{-m}$, the water permeability is $k_{rw} = S_w^4$ (Leong and Rahardjo, 1997) and the air permeability follows Brooks and Corey with $\lambda = 3$, $k_{ra} = (1 - S_w)^2(1 - S_w^{5/3}) \geq k_{ra}^{\min} = 10^{-4}$.

The effective stress follows the Mohr–Coulomb criterion with elastic moduli E, ν. With an associated flow rule ($\psi = \phi'$) the return mapping is analytical, with the consistent tangent; with $\psi < \phi'$ an iterative non-associated return mapping is used. The strength envelope of Fredlund et al. (1978) with ϕ^b corresponds to $\chi = \tan\phi^b / \tan\phi'$ on the suction; Rahardjo et al. (2007) use $\phi^b = \phi'$, i.e. $\chi = 1$, which is the default here; $\chi = S_e$ is the alternative examined in Section 8.

Four models are compared (Table 1). In the uncoupled models the terms $S_w\dot{\varepsilon}_v$ and $S_a\dot{\varepsilon}_v$ are removed from Eq. (3), i.e. the flow is solved on a rigid skeleton, as in a seepage code; the equilibrium still uses the pressures. In the passive-air models $p_a = 0$ in the whole domain and the air mass balance is not solved. Model (iii) gathers the hypotheses of Rahardjo et al. (2007) and of most of the literature.

Table 1: Models compared.

model	coupling	air
(i) complete	coupled	active ($p_a = 0$ only on the ground surface)
(ii)	coupled	passive ($p_a = 0$)
(iii) the usual one	uncoupled	passive
(iv)	uncoupled	active (two-phase flow on a rigid skeleton)

2.2. Boundary conditions

The rainfall is a flux on the horizontal projection of the ground surface with the unilateral condition $p_w \leq 0$, inflow $\leq I_r$ and complementarity, solved as an active set of ponded vertices; without rainfall the same condition is a seepage face. The pore air is atmospheric on the ground surface and the base and lateral boundaries are closed to the air (the effect of opening them is examined in Section 6.1). The water pressure is hydrostatic on the lateral boundaries below the water table and the boundaries are impervious above it. The base is fixed and the lateral boundaries are on rollers.

2.3. Discretization and solution

Displacements are interpolated by quadratic and pressures by linear triangles (Taylor–Hood), in plane strain. The storage terms in $\dot{S}_w$ and $\dot{p}_a$ are lumped at the vertices in the mass-conservative secant form $n(S^{n+1} - S^n)$ (Celia et al., 1990). The equations are integrated by the implicit Euler method and solved by Newton's method with the complete consistent Jacobian, including the derivatives of the permeabilities, of the saturation and of χ; Dirichlet conditions are eliminated and the linear systems are solved with Pardiso. A time step that does not converge is restarted from its initial state with up to five levels of bisection. The time step is 15 min during the rainfall, 1 h in the following 24 h and 4 h afterwards.

2.4. Factor of safety

At selected instants an independent stability analysis is made with p_w and p_a frozen, as in the uncoupled practice: from the geostatic state with the strength increased (c'/F_0, $\tan\phi'/F_0$ with $F_0 = 0.6$) the strength reduction factor F is increased until the equilibrium no longer converges (Zienkiewicz et al., 1975; Griffiths and Lane, 1999); the FS is the last converged F, found by bisection with a tolerance of 0.002 in the cases near the critical intensity and 0.01 in the others. The state of the transient is restored afterwards. The strength reduction is always made with $\psi = \phi'$: with $\psi < \phi'$ it does not converge reliably (Tschuchnigg et al., 2015). In the coupled transient, the loss of equilibrium with the real strength (the transient does not converge even with five bisections) is recorded as the time t_f; the transient then continues with the strength increased, only to follow the pore pressures.

3. Verification and validation

3.1. Drainage of a sand column (Liakopoulos test)

In the test of Liakopoulos (1965) a 1-m column of Del Monte sand, initially saturated by a steady downward flow with $p_w = 0$ throughout, drains through its base after the inflow at the top is stopped. The test is a standard benchmark of coupled two-phase formulations (Lewis and Schrefler, 1998; Schrefler and Scotta, 2001; Khoei and Mohammadnejad, 2011) and its measured outflow and pore-water pressures are available (Callari and Abati, 2009). The properties are those of Lewis and Schrefler (1998): $E = 1.3$ MPa, $\nu = 0.4$, $n = 0.2975$, intrinsic permeability 4.5×10^{-13} m^2 ($k_s = 4.41 \times 10^{-6}$ m/s), $S_w = 1 - 0.10152\,(p_c/\gamma_w)^{2.4279}$ (p_c/γ_w in metres), $k_{rw} = 1 - 2.207\,(1 - S_w)^{1.0121}$ and the Brooks–Corey air permeability with $S_r = 0.2$; $\chi = S_w$, elastic soil. The air pressure is atmospheric at both ends, the water pressure at the base, and the top is impervious to water. The column was discretized with 2×80 cells of two triangles and time steps of 30 s.

The two-phase solution agrees with that of Khoei and Mohammadnejad (2011) (Fig. 1a,b; Table 2): the water pressure at the top differs by 0.43 kPa (7%) after 5 min and by less than 0.25 kPa afterwards, the minimum air pressure, about −5.5 kPa near the top where the air enters the drying sand, by less than 0.3 kPa, and the settlement of the top by less than 0.07 mm. The passive-air solution reproduces the measured outflow and pore pressures (Fig. 1c,d), as other single-phase solutions do (Callari and Abati, 2009). The two-phase model, with the usual Brooks–Corey air permeability, drains more slowly than the test; the same happens in the two-phase solution of Khoei and Mohammadnejad (2011), so this is a feature of the air permeability law and not of the implementation. The test verifies the

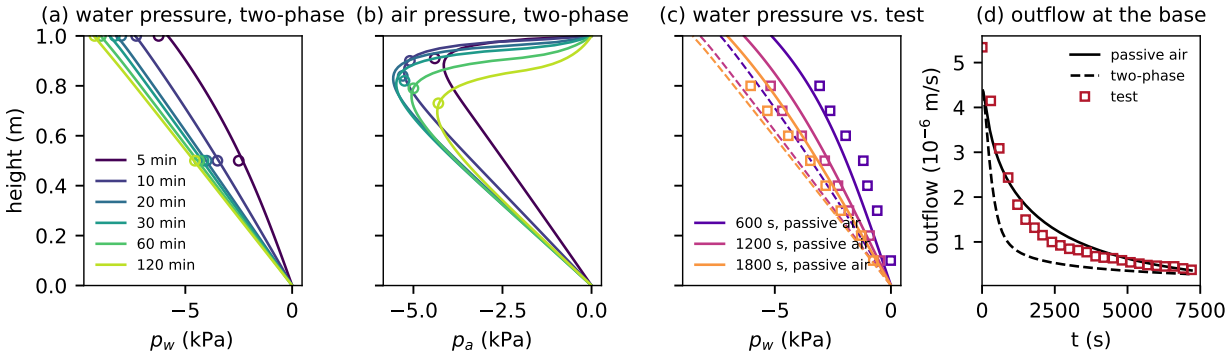


Figure 1: Drainage of the Liakopoulos column. (a) Water and (b) air pressure profiles of the two-phase model, with the solution of Khoei and Mohammadnejad (2011) (circles; water pressure at the top and at 0.5 m, minimum air pressure). (c) Water pressure of the passive-air model (solid) and of the two-phase model (dashed) and the measured values (squares; Callari and Abati, 2009). (d) Outflow at the base and the measured outflow.

Table 2: Liakopoulos test, two-phase model: NeoPZ and Khoei and Mohammadnejad (2011) (values read from their Fig. 1, uncertainty about 0.15 kPa and 0.03 mm).

t	p_w at the top (kPa)		p_w at 0.5 m (kPa)		min. p_a (kPa)		settlement (mm)	
(min)	NeoPZ	ref.	NeoPZ	ref.	NeoPZ	ref.	NeoPZ	ref.
5	-5.82	-6.25	-2.32	-2.50	-4.15	-4.40	0.90	0.97
10	-7.35	-7.30	-3.44	-3.50	-5.26	-5.10	1.26	1.26
20	-8.22	-8.00	-4.03	-4.00	-5.57	-5.30	1.45	1.40
30	-8.54	-8.30	-4.21	-4.15	-5.48	-5.25	1.51	1.46
60	-8.96	-8.95	-4.43	-4.35	-5.05	-5.00	1.58	1.55
120	-9.29	-9.25	-4.60	-4.55	-4.33	-4.30	1.62	1.60

air phase and the coupling, and shows that, even in drainage, the air resistance reduces the outflow by a factor of two to three between 10 and 30 min.

3.2. Reference slope

The slope of Rahardjo et al. (2007) is 10 m high, inclined at 45°, with 30-m platforms at the crest and at the toe and the base 20 m below the toe; the initial water table is 5 m below the toe and inclined at 7°, and the initial suction is hydrostatic and limited to 75 kPa (Table 3, Fig. 2). The standard mesh has 1684 triangles, with elements of 0.75 m down to 2 m below the surface.

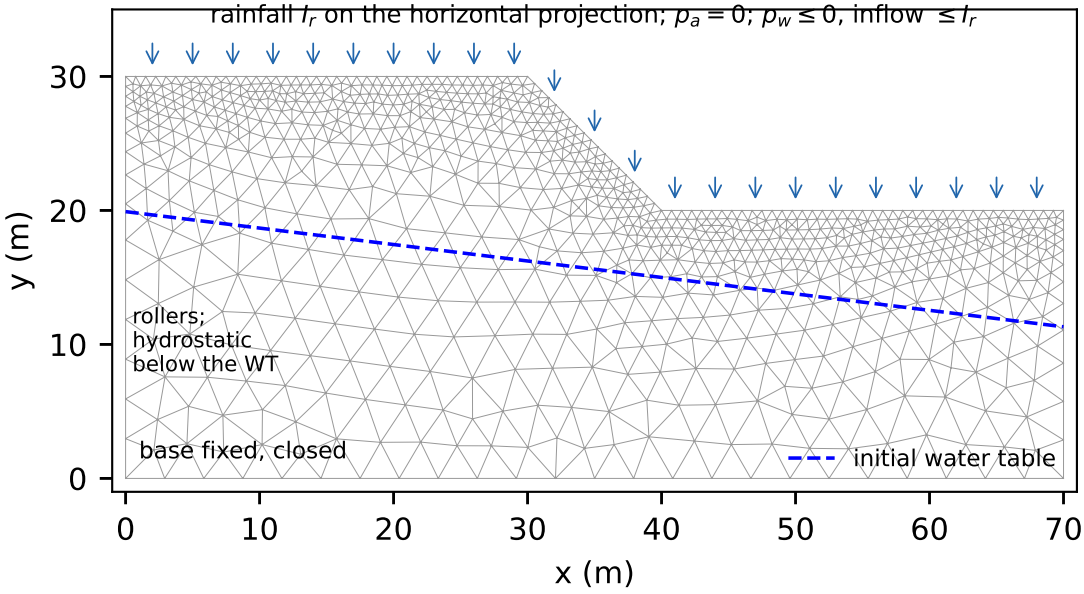


Figure 2: Reference slope (Rahardjo et al., 2007): geometry, standard mesh, initial water table and boundary conditions.

The initial FS agrees with that of Rahardjo et al. (2007): 2.119 against 2.11 for the reference slope and differences of at most 0.05 for heights of 5 to 40 m and water tables 2.5 to 15 m deep; a simplified Bishop analysis with a search of circular surfaces on the same initial pressures gives 2.092. With model (iii) the minimum FS under 24-h rainfalls follows the sigmoid fitted by Rahardjo et al. (2007) to their results (their Eq. 6) within 0.01–0.14, the largest differences in the transition between the upper and the lower plateau. The static benchmark of Zhang et al. (2014), which compares six laws for the suction strength, is reproduced within 0–5%.

4. Failure criterion and critical intensity

The slope fails when the FS reaches one. Since the FS of each instant is computed with the pressures of that instant frozen, a value below one does not describe a physical state: the slope would already have moved and redistributed the pressures. Below one the FS only measures how far equilibrium has been exceeded. The quantity to compare is therefore

Table 3: Parameters of the reference slope (Rahardjo et al., 2007); E, ν, ψ and the air parameters are assumptions of this study.

soils f_{10}, f_{50}, f_{100}	$a = 10$, 50, 100 kPa; $k_s = 10^{-4}$, 10^{-5}, 10^{-6} m/s (360, 36 and 3.6 mm/h)
retention, k_{rw}	Fredlund–Xing, $n = m = 1$, $\theta_s = 0.45$; $k_w = k_s S_w^4$
strength	$c' = 10$ kPa, $\phi' = 26°$, $\chi = 1$ ($\phi^b = \phi'$), $\gamma = 20$ kN/m^3
elasticity, dilatancy	$E = 20$ MPa, $\nu = 0.3$; $\psi = \phi'$ (associated) or $\psi = 0$
air	Brooks–Corey $\lambda = 3$, $k_{ra}^{\min} = 10^{-4}$, $\mu_w/\mu_a = 55.6$
initial state	water table 5 m below the toe, inclined at 7°; suction ≤ 75 kPa
rainfall	uniform, 24 h; 48 h simulated

whether, with which rainfall and at which instant, the FS reaches one. We define the critical intensity I_c as the smallest intensity of a 24-h rainfall whose minimum FS is ≤ 1.01, obtained by linear interpolation between the computed intensities that bracket it, and the time to failure as the first instant with FS ≤ 1.01 (or t_f, if the coupled transient loses equilibrium earlier). The threshold 1.01 accounts for the resolution of the FS: the bisection tolerance (0.002) and the discretization error, about 0.01 for the standard mesh (Section 6.2).

The lower FS plateau of the intense rainfalls, the value that the usual analyses report as the minimum FS, is the FS of the slope saturated up to its surface with the water flowing through it. A single steady-state calculation with the surface ponded gives 0.84–0.85 for the three soils, the plateau of model (iii); it is reached after about 7, 24 and 170 h of ponding in the f_{10}, f_{50} and f_{100} soils, respectively, and depends on the geometry and on the orientation of an anisotropic conductivity tensor (0.63–1.05), not on the magnitude of k_s. The value 1.45 of Rahardjo et al. (2007) for the f_{100} soil is the FS at the end of 24 h of rainfall, not a plateau. A plateau below one distinguishes no slopes that have already failed; it only matters when it stays above one, in which case no rainfall of that duration fails the slope.

5. Critical intensity of the reference slope

Figure 3 and Table 4 give the minimum FS under 24-h rainfalls of the f_{50} soil ($k_s = 36$ mm/h) in the four models, the critical intensities and the times to failure. All the models have a critical intensity: from I_c on, the FS reaches one during the rainfall and does not rise again while it rains (Fig. 6). In this sense the limiting intensity of Rahardjo et al. (2007) exists in every model. What changes is its value. The usual model (iii) gives $I_c = 48$ mm/h ($1.3k_s$); the complete model (i) gives 31 mm/h ($0.9k_s$) with an associated flow rule and 26 mm/h ($0.7k_s$) without dilatancy. The hypotheses of the usual chain overestimate by 54 to 86% the rainfall that brings the slope to failure; against the sigmoid published by Rahardjo et al. (2007), which reaches one at about 40 mm/h, by 28 to 54%. They also delay the failure: with 54 mm/h the FS of model (iii) reaches one after 22 h, against 18 h in model (i) with $\psi = \phi'$ and the loss of equilibrium after 12.5 h with $\psi = 0$; with 90 mm/h, after 20, 18 and 12 h.

The four models separate the two hypotheses (Table 4).

- *The air lowers the critical intensity by about half*: on a rigid skeleton from 48 to

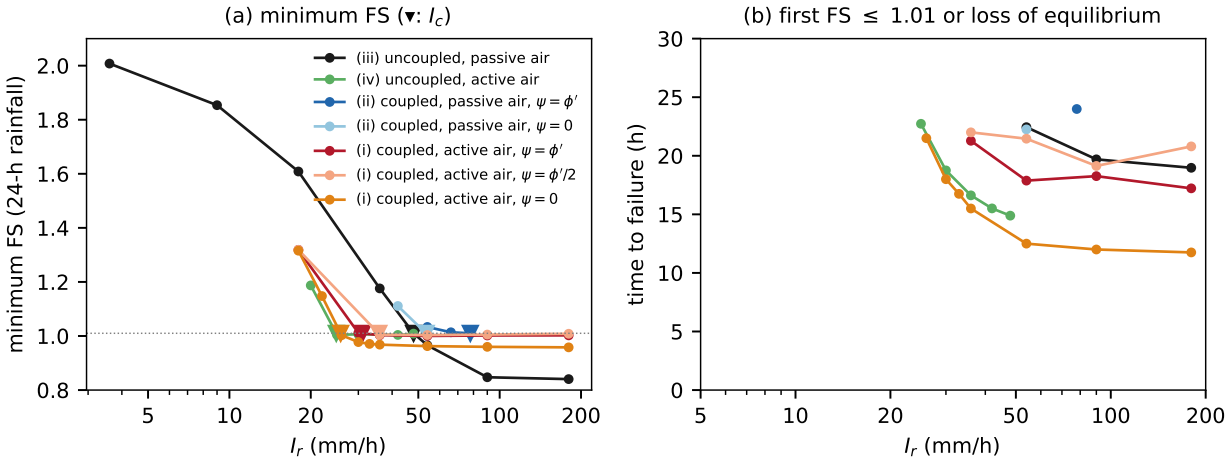


Figure 3: Soil f_{50}, $\alpha = 45°$, 24-h rainfalls. (a) Minimum FS against the rainfall intensity in the four models (▼: critical intensity, FS = 1.01). (b) Time to failure: first instant with FS ≤ 1.01, or loss of equilibrium of the coupled transient with $\psi = 0$.

Table 4: Critical intensity ($k_s = 36$ mm/h) and time to failure of the reference slope. Bracket: computed intensities on each side of I_c (FS ≤ 1.01); the column FS ≤ 1.02 gives I_c with that threshold. Time to failure: first instant with FS ≤ 1.01; –: no failure; blank: not computed. †Loss of equilibrium of the coupled transient.

model	I_c (mm/h)	bracket (mm/h)	I_c/k_s	I_c FS ≤ 1.02	time to failure (h) for I_r (mm/h) = 36	48	54	90
(iii) uncoupled, passive air	48	48–54	1.33	47	–	–	22.4	19.7
(iv) uncoupled, active air	25	20–25	0.69	25	16.6	14.9		
(ii) coupled, passive, $\psi = \phi'$	78	66–78	2.16	62			–	
(ii) coupled, passive, $\psi = 0$	53	42–54	1.49	52			22.2†	
(i) coupled, active, $\psi = \phi'$	31	30–36	0.86	30	21.3		17.9	18.3
(i) coupled, active, $\psi = \phi'/2$	36	18–36	0.99	35	22.0		21.5	19.1
(i) coupled, active, $\psi = 0$	26	22–26	0.72	26	15.5†		12.5†	12.0†

25 mm/h (passive and active air, models iii and iv); in the coupled models from 53 to 26 mm/h without dilatancy and from 78 to 31 mm/h with $\psi = \phi'$ (from 62 to 30 mm/h with the threshold 1.02).

- *The coupling raises it*: by 4–10% without dilatancy (from 25 to 26 mm/h with active air, from 48 to 53 mm/h with passive air) and by 24–63% with associated dilatancy (to 31 and 78 mm/h). The deformation of the skeleton during the wetting, elastic swelling as the effective stress decreases and plastic dilation in the shear zone, opens pore space and lowers the pore pressures.

The two simplifications of the usual model thus have opposite effects: neglecting the deformation is conservative, neglecting the air is not, and the air dominates. A two-phase flow analysis on a rigid skeleton, model (iv), recovers the bias: its critical intensity, 25 mm/h, is 4% below that of the complete model without dilatancy and 20% below that with associated dilatancy, and its time to failure with 36 mm/h, 16.6 h, lies between those of the complete model with $\psi = 0$ (15.5 h) and with $\psi = \phi'$ (21.3 h).

The coupled models with an associated flow rule approach one from above (1.000–1.014 for $I_r \geq I_c$), so their critical intensity depends on the threshold. With FS ≤ 1.02 instead of 1.01 the critical intensities change by less than 1 mm/h, except that of model (ii) with $\psi = \phi'$, whose minimum FS is 1.033, 1.014 and 1.010 with 54, 66 and 78 mm/h: $I_c = 62$ instead of 78 mm/h. In this model the slope sits in limit equilibrium over a wide range of intensities, with plastic deformation but without loss of equilibrium.

6. Trapped air

Rahardjo et al. (2007) classified the slopes with $k_s \leq 10^{-6}$ m/s as safe in rainfalls of up to 24 h, whatever the intensity, because the rain does not raise the water table. We examine the f_{100} soil ($k_s = 3.6\,\mathrm{mm/h}$) on the steepest slope of their study, $\alpha = 63.4°$, whose initial FS is 1.775 (Fig. 4). With the usual model (iii) the minimum FS under 24-h rainfalls is 1.38, 1.16, 1.10 and 1.09 for $I_r = 9$, 18, 36 and 80 mm/h: the slope does not fail, as the classification states. With model (ii) the FS is 0.02–0.04 higher. With active air it is 1.17, 1.03, 1.02 and 1.01: from $I_r = 5k_s$ on, the slope reaches limit equilibrium at the end of the rainfall.

The mechanism is not more water. With active air 10.4–11.6 $\mathrm{m^3/m}$ infiltrate in 24 h, against 13.5–17.9 $\mathrm{m^3/m}$ with passive air (Fig. 4b). The air resistance slows the infiltration and the surface ponds earlier: with 9 mm/h the first vertex ponds after 6 h with active air and after 11.75 h with passive air, and at the end of the rainfall 83 and 41 of the 96 surface vertices are ponded; with 36 mm/h the whole surface ponds within 4.5 h. Once the surface is sealed, the air between the wetting front and the water table can no longer escape and is compressed by the water that still enters: with 36 mm/h p_a reaches 19.5 kPa after 1 h, 30 kPa after 3 h and 38 kPa after 24 h, and averages 32 kPa between 0.5 and 5 m below the surface. The matric suction does not decrease, it increases: between 2 and 5 m it is 38 kPa with active air and 24 kPa with passive air (Fig. 4d), because less water has arrived. What rises is the water pressure, pushed by the compressed air: -6 against $-24\,\mathrm{kPa}$ in the same layer (Fig. 4c). With $\chi = 1$ the effective stress depends only on p_w, and the shear strength decreases although the soil is drier. The usual rule that the stability depends on

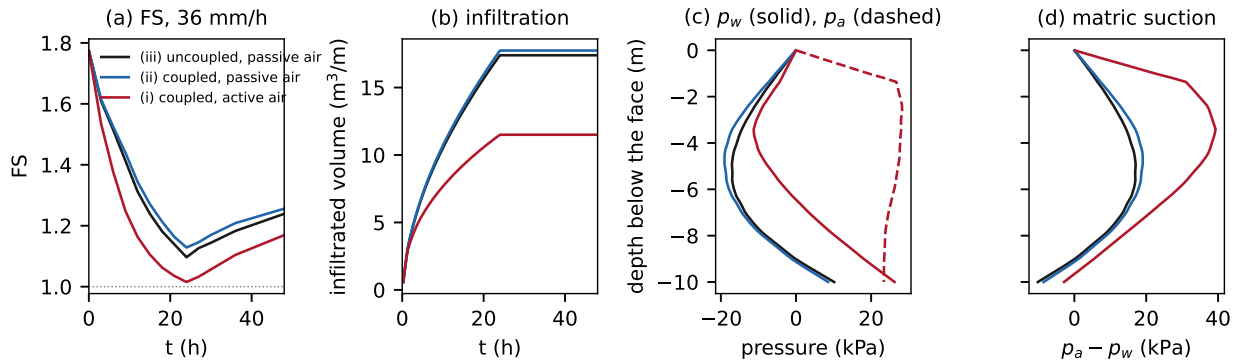


Figure 4: Soil f_{100}, $\alpha = 63.4°$, $I_r = 36\,\mathrm{mm/h}$ during 24 h, in models (i)–(iii). (a) FS; (b) infiltrated volume; (c) water (solid) and air (dashed, model i) pressures and (d) matric suction along a vertical below the middle of the face at $t = 24$ h.

the infiltrated volume therefore compares cases of the same model only; between models, less water gives a smaller FS.

The air pressure is controlled by how easily the air leaves through the ponded surface. With the minimum relative permeability of the air $k_{ra}^{\min}$ between 10^{-6} and 10^{-3} the minimum FS (1.013–1.015) and the maximum air pressure (37–38 kPa) do not change; with 10^{-2} the air escapes through the ponded surface, p_a falls to 28 kPa and the FS rises to 1.05.

6.1. Confinement of the air

The base and the lateral boundaries are closed to the air in the reference model, as in the other studies of rainfall infiltration with the air phase. This is the upper bound of the entrapment through the boundaries. Opening them (atmospheric air pressure on the base and on the lateral boundaries) changes the minimum FS by less than 0.002: 1.017 against 1.015 for the f_{100} slope with 36 mm/h, and 1.011 against 1.011 for the f_{50} slope with 30 mm/h. The lateral boundaries are 30 m away from the slope and the base is below the water table; the air is sealed by the ponded surface above it, not by the boundaries of the domain.

6.2. Mesh

The minimum FS of the f_{100} slope with 36 mm/h decreases with the mesh: 1.0154, 1.0122 and 1.0090 with elements of 0.75, 0.5 and 0.35 m near the surface (1764, 3814 and 7622 triangles), without a clear sign of convergence, while the initial FS converges (1.775, 1.761, 1.752 and 1.748 with 0.25 m; Fig. 5). The margin of 1.5% of the standard mesh is of the order of the discretization error, and by the criterion adopted the case is imminent failure; whether the converged minimum FS falls below one would require finer meshes or a resolution of the shear band.

7. Dilatancy and the response after failure

The dilatancy does not decide whether a critical intensity exists, but it moves it and it decides what happens afterwards (Figs. 6 and 3). Without dilatancy ($\psi = 0$) the coupled transient loses equilibrium for $I_r \geq 26\,\mathrm{mm/h}$, between 21.5 h (26 mm/h) and 11.75 h

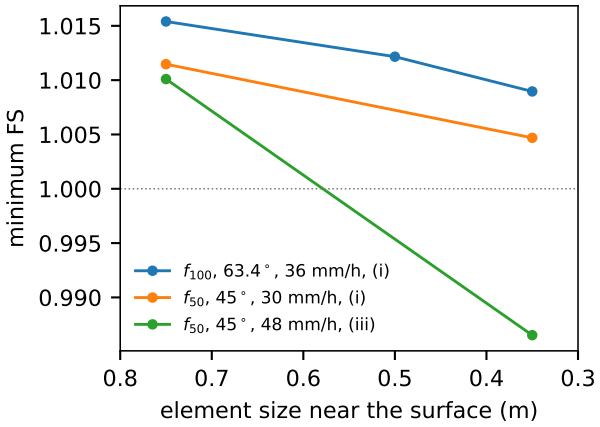


Figure 5: Minimum FS against the element size near the surface: f_{100} slope at 63.4° with 36 mm/h in model (i), and the f_{50} reference slope near its critical intensities in models (i) and (iii).

(180 mm/h); with 22 mm/h it does not. With $\psi = \phi'/2$ or $\psi = \phi'$ the FS stops at 1.000–1.008 and the transient never loses equilibrium: the slope sits in limit equilibrium, with plastic deformation and the suction generated by the plastic dilation of the shear zone. Half of the associated dilatancy is enough for this behaviour. Below the critical intensity the flow rule hardly matters: with 18 mm/h the minimum FS is 1.316 with $\psi = 0$ and 1.319 with $\psi = \phi'/2$ and ϕ'. The dilatancy shifts I_c by about 20% (from 26 to 31 mm/h) and brings the failure forward by about 6 h in the intense rainfalls (Table 4).

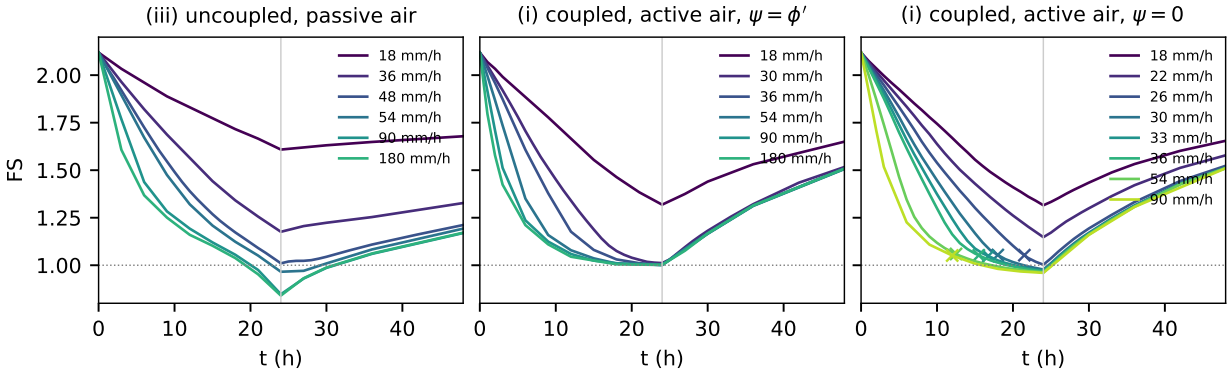


Figure 6: FS during and after 24-h rainfalls of several intensities, soil f_{50}, $\alpha = 45°$: model (iii) and model (i) with $\psi = \phi'$ and $\psi = 0$ (×: loss of equilibrium of the coupled transient; afterwards the transient continues with the strength increased, only to follow the pressures).

With $\psi = 0$ the loss of equilibrium occurs when the FS of the strength reduction, computed with $\psi = \phi'$, is still 1.05–1.06. Either the associated strength reduction overestimates by about 5% the FS of a soil without dilatancy, or part of the loss of equilibrium is numerical. The time step does not explain it: with a time step of 0.125 h instead of 0.25 h during the rainfall, the transient with 36 mm/h loses equilibrium after 15.6 h instead of 15.5 h. For $\psi = 0$ the time of failure is therefore t_f and $22 < I_c \leq 26\,\mathrm{mm/h}$; the FS values below one after t_f use the pressures of a transient continued with the strength increased and only show

that the equilibrium was exceeded.

The flow rule matters as much without active air. In model (ii), coupled with passive air, the transient without dilatancy loses equilibrium after 22.3 h with 54 mm/h and the critical intensity is 53 mm/h; with $\psi = \phi'$ the minimum FS stays between 1.010 and 1.033 from 54 to 78 mm/h and the transient never loses equilibrium. Dilatancy, through the suction it generates in the shear zone, is the main reason why the coupled models with an associated flow rule do not predict a collapse, and the reason why the plateau of the coupled models is close to one instead of the 0.84–0.85 of the saturated slope.

8. Generalization

Table 5 extends the comparison of models (i) and (iii) to the permeable f_{10} soil, to the Bishop parameter $\chi = S_e$ and to the low-permeability f_{100} soil on two slopes.

Table 5: Critical intensity (mm/h) of 24-h rainfalls in the usual model (iii) and in the complete model (i, $\psi = \phi'$). >: no failure up to the most intense rainfall computed, whose minimum FS is given in parentheses; *limit equilibrium (minimum FS ≤ 1.02).

slope and soil	k_s (mm/h)	I_c, model (iii)	I_c, model (i)
f_{50}, $\alpha = 45°$, $\chi = 1$ (reference)	36	48	31
f_{10}, $\alpha = 45°$, $\chi = 1$	360	135	139
f_{50}, $\alpha = 45°$, $\chi = S_e$	36	48	30
f_{100}, $\alpha = 63.4°$, $\chi = 1$	3.6	> 80 (1.093)	> 80 (1.013)*
f_{100}, $\alpha = 45°$, $H_s = 20$ m, $\chi = 1$	3.6	> 80 (1.252)	> 80 (1.172)

Permeable soil. In the f_{10} soil ($k_s = 360\,\text{mm/h}$) the critical intensity is 135 mm/h in model (iii) and 139 mm/h in model (i): the bias of the usual model disappears. The critical intensities are well below k_s ($0.4k_s$), the surface does not pond before the slope fails and the air escapes through it. The bias is a feature of the regime $I_c \gtrsim k_s$, in which the surface seals the air: in the f_{50} soil I_c of model (iii) is $1.3k_s$.

Low-permeability soil. In the f_{100} soil ($k_s = 3.6\,\text{mm/h}$) no 24-h rainfall of up to 80 mm/h brings a 20-m, 45° slope to failure in either model; the complete model gives a minimum FS 0.08–0.11 lower (1.17 against 1.25 with 80 mm/h), and its minimum occurs after the rainfall, 30–36 h after its start. On the 63.4° slope the usual model stays 9–10% above failure at any intensity, whereas the complete model reaches limit equilibrium (Section 6). For this soil the comparison is not a shift of I_c but the difference between a safe slope and one at the limit.

Bishop parameter. With $\chi = S_e$ (for the Fredlund–Xing curve with $C(\psi) = 1$, $S_e = S_w$) the suction contributes less to the initial strength (initial FS 1.895 instead of 2.119) and the air pressure also enters the effective stress, through $(1 - \chi)p_a$. The critical intensity of model (iii) is unchanged, 48 mm/h (minimum FS 1.010 and 0.998 with 48 and 54 mm/h), and that of model (i) falls from 31 to 30 mm/h: the bias persists (+62%). Near failure the controlling zone is nearly saturated and $\chi \to 1$ in both laws, so the law of χ changes the initial FS much more than the critical intensity.

Confinement of the air. Opening the base and the lateral boundaries to the air changes the minimum FS by less than 0.002 (Section 6.1): the air is sealed by the ponded surface, not by the boundaries of the domain.

Mesh. With elements of 0.35 m instead of 0.75 m near the surface (8052 elements instead of 1834; initial FS 2.106 instead of 2.119), the minimum FS of model (i) with 30 mm/h falls from 1.011 to 1.005 and that of model (iii) with 48 mm/h from 1.010 to 0.987 (Fig. 5). With the local slope of the minimum FS with the intensity these differences shift the critical intensities to about 30 and 46 mm/h; the ratio between them, 1.5, does not change.

9. Implications for rainfall thresholds

Physically based rainfall thresholds are the critical intensities of the analyses, for a set of durations. For the reference slope, the usual model places the 24-h threshold at $1.3k_s$; the complete model at $0.9k_s$ with an associated flow rule and $0.7k_s$ without dilatancy. A threshold derived from the usual chain is thus not conservative: rainfalls between the two thresholds, which the usual analysis classifies as safe, bring the slope to failure. In a probabilistic setting, where the annual failure probability integrates the conditional probability of failure over the intensity–frequency distribution of the rainfall (e.g. Tang et al., 2018), the shift of the threshold changes the exceedance probability of the failure-triggering rainfall by a factor that depends on the local intensity–duration–frequency curve; for a 24-h rainfall, a threshold 35–46% lower corresponds to a much more frequent event.

Two of the conclusions of Rahardjo et al. (2007) change with the failure criterion and the complete model. (1) Low permeability does not guarantee safety under short rainfalls: the trapped air brings the f_{100} slope at 63.4° to limit equilibrium in 24 h, a slope that the usual analysis places 9–10% above failure. (2) The limiting intensity and the lower plateau are not properties of the soil: the plateau is the FS of the slope saturated to its surface and can be computed directly, with a single steady-state calculation with the surface ponded, without simulating the rainfall; whether the slope fails depends on whether that value is below one.

When is the usual model enough? The bias comes from the air sealed below a ponded surface. For rainfall intensities well below k_s the surface does not pond before the slope fails, the air escapes and the usual model gives the critical intensity of the complete one (135 and 139 mm/h for the f_{10} soil). The bias concentrates in the regime $I_c \gtrsim k_s$, in which the surface ponds early, which is the regime of the critical intensities of the f_{50} and f_{100} soils. In that regime a two-phase flow analysis on a rigid skeleton recovers the bias and errs on the safe side by 4–20% (model iv); the usual single-phase seepage analysis does not.

10. Limitations

- Two-dimensional, homogeneous slope with a single geometry for the decomposition; the critical intensity of other slopes was examined only for two soils, two Bishop laws and one steeper slope.

- The air leaves only through the ground surface, and once it ponds only through the minimum relative permeability $k_{ra}^{\min}$. Opening the base and the lateral boundaries does not change the results (Section 6.1), but real slopes have macropores, cracks, vegetation and heterogeneities that let the air out through a ponded surface; their effect is represented here only by $k_{ra}^{\min}$, and with $k_{ra}^{\min} = 10^{-2}$ the FS rises by 0.035.

- The strength reduction is associated even when the transient uses $\psi = 0$; the loss of equilibrium of the transient is then the failure indicator, and part of it may be numerical.

- No hysteresis of the retention curve, constant porosity in the hydraulic coefficients, no dissolved air, no vapour and an isothermal ideal gas.

- The mesh error of the FS was assessed in three cases (Fig. 5): with 0.35-m elements the minimum FS falls by 0.007–0.024 and, in the f_{100} case, by about 0.003 per refinement without clear convergence; the critical intensities shift by 1–2 mm/h.

- The air phase was validated in drainage (Liakopoulos), not in infiltration with air entrapment; a validation against a column with measured air pressure under infiltration (Touma and Vauclin, 1986) is still needed.

11. Conclusions

A coupled formulation with displacement, pore-water and pore-air pressures was implemented in NeoPZ, validated against the Liakopoulos drainage test and used to quantify how the hypotheses of the usual chain of analysis, uncoupled seepage with atmospheric pore air followed by a stability analysis with the frozen pressures, bias the rainfall that brings an unsaturated slope to failure. The main conclusions are the following.

1. With the failure criterion FS ≤ 1, models must be compared by the critical intensity and the time to failure, not by the minimum FS. A minimum FS below one is the FS of the slope saturated to its surface, a property of the boundary-value problem that can be computed directly with the surface ponded.
2. For the homogeneous slope of Rahardjo et al. (2007) with $k_s = 36\,\mathrm{mm/h}$ the usual model gives a critical intensity of 48 mm/h ($1.3k_s$) for 24-h rainfalls, and the complete model 31 mm/h ($0.9k_s$) with an associated flow rule and 26 mm/h ($0.7k_s$) without dilatancy. The usual hypotheses overestimate the failure-triggering rainfall by 54–86% and delay the predicted failure by up to 10 h in intense rainfalls. The bias persists with $\chi = S_e$ (48 against 30 mm/h) and with a finer mesh (about 46 against 30 mm/h).
3. The two simplifications act in opposite directions. The trapped air halves the critical intensity; the coupling raises it, by 4–10% through the elastic swelling of the skeleton and by up to 63% when the plastic dilation of an associated flow rule is added. The air dominates, and a two-phase flow analysis on a rigid skeleton recovers the bias with a critical intensity 4–20% below that of the complete model.
4. The trapped air lowers the FS with less water. Once the surface ponds, the air between the wetting front and the water table is compressed to 36–38 kPa; less water infiltrates and the matric suction increases, but the pore-water pressure rises with the air pressure and the effective stress falls. A low-permeability slope ($k_s = 3.6\,\mathrm{mm/h}$, 63.4°) that the usual model places 9–10% above failure reaches limit equilibrium; the effect does not depend on whether the base and lateral boundaries are open to the air.
5. Dilatancy does not decide whether a critical intensity exists but shifts it by about 20% and decides the response after it is exceeded: without dilatancy the coupled transient

loses equilibrium, independently of the time step; with $\psi \geq \phi'/2$ the slope stays at limit equilibrium.

6. The bias concentrates in the regime $I_c \gtrsim k_s$, in which the surface ponds and seals the air. For a permeable soil whose critical intensity is well below k_s the usual model is adequate (135 against 139 mm/h).

Rainfall thresholds derived from uncoupled, passive-air analyses are therefore unconservative for slopes whose critical rainfall exceeds the saturated conductivity of the surface soil. The validation of the air phase in infiltration with measured air pressures, natural (open, heterogeneous) slopes and three-dimensional effects are the next steps.

Code and data availability

The material `TPZMatUnsaturatedUPwPa`, the program `SlopeRainfall` and the scripts that run all the cases and draw the figures are part of the NeoPZ library (`https://github.com/labmec/neopz`); the results are in `Projects/SlopeRainfall/resultados`.

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